In-orbit commissioning of nanosatellite done successfully

5G satellite operator OQ Technology has successfully completed the in-orbit commissioning of its Tiger-2 nanosatellite. During tests, OQ was able to send the terminal's GPS location to the satellite from inside a fast-moving car without having a direct line of sight to the sky. Even when buried in the desert sand, the terminal was still sending signals to the satellite, making it ideal for many agricultural applications. The high level of signal-to-noise ratio surpassed OQ's high expectations. OQ Technology is planning to launch a constellation of 72 satellites, providing 5G IoT and machine-to-machine communication as well as ultra-reliable low latency communication.

OQ terminal left with battery in the desert successfully delivered data to its Tiger-2 satellite over multiple weeks (Credit: www.oqtec.space)

New sensor for tracking nanoparticles' movement in space

Nanoparticles are omnipresent in our environment—from viruses in ambient air and proteins in the body to building blocks of new materials for electronics and in surface coatings. Visualising such small particles is a problem as they can hardly be seen under an optical microscope. Now, researchers of Karlsruhe Institute of Technology have developed a sensor that not only detects nanoparticles but also determines their condition and tracks their movements in space. In this way, the sensor might provide insights into not yet understood processes, particularly related to human biology.

Physicist Larissa Kohler, KIT, has developed a new type of resonator that makes ever smaller nanoparticles visible (Credit: www.kit.edu/kit)

Innovative fabric enables battery-free communication

Imagine your car starting the moment you get in because it recognises the jacket you are wearing. Consider the value of a hospital gown that continuously measures and transmits a patient's vital signs. These are just two applications made possible by a new 'body area network-enabling' fabric invented by engineers at the Henry Samueli School of Engineering, University of California, Irvine. The invention allows the creation of flexible textiles to create a system capable of battery-free communication between articles of clothing and nearby devices. This also enables wearers to digitally interact with nearby electronic devices and make secure payments with a single touch or swipe of a sleeve.

An innovative, new high-tech fabric created by engineers at UCI enables wearers to communicate with others, wirelessly charge devices and pass through security gates with the wave of an arm (Credit: https://news.uci.edu)